

NITDGP/BTECH/Reg/Even/2025-26

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
Even Semester End-term Examination, 2025-26

Course Code: BTC401

Course Name: Molecular Biology and Genetic Engineering

Full Marks: 60

Time: 3 h

Instructions: Answer **ALL** the questions.

Q. No.	Questions	Marks	Mapped CO
1	What is semiconservative mode of replication? How Meselson-Stahl proved semiconservative mode of DNA replication? Define the activity of RNase H.	1+3+1	CO1
2	Differentiate between fidelity and processivity of DNA polymerase. Depict the process of mismatch repair and discuss how it contributes to the high fidelity of DNA Pol <i>in vivo</i> .	2+3	CO1
3	Discuss the role of RNA Pol CTD phosphorylation in mRNA synthesis. Explain how transcription termination occurs in eukaryotic cells?	2+3	CO1
4	What could be the consequence of a mutation at the -10 element of a prokaryotic promoter? Discuss two differences between the prokaryotic and eukaryotic transcription. What is RdRP and where usually we find it?	2+2+1	CO3
5	What do you understand by a housekeeping gene? Give one example. Discuss how gene expression can be regulated positively or negatively? You have started an <i>E. coli</i> culture in the media which contains glucose and lactose. Will you get activation of <i>lac</i> operon? Justify your answer.	1+2+2	CO1
6	Discuss how DNA methylation regulates gene expression. State how different steps in gene expression pathway that can be regulated to affect gene expression (discuss any four steps).	3+2	CO1
7	Discuss the principle of alpha complementation with neat diagrams. What is its application in DNA cloning? How is the copy number of a plasmid regulated and why is it important?	2+1+2	CO2
8	Discuss YAC and mammalian expression vectors with appropriate diagrams.	2.5+2.5	CO2
9	You are asked to digest your vector with <i>EcoRI</i> (site 5'G↓AATTC3') and insert with <i>SmaI</i> (site 5'CCC↓GGG3'). You are also provided with Ligase. Will you be able to efficiently ligate the digested vector and insert? Do you need any additional reagents to complete the cloning? Justify your answers. What is isoschizomers?	4+1	CO4
10	What is a cDNA library and how is it different from a genomic DNA library? Illustrate the method of cDNA library preparation using a flowchart.	2+3	CO2
11	a) Discuss the different ways 1 st strand cDNA synthesis can be primed. b) A scientist wanted to study role of different genes in lung cancer. She has taken a normal lung cell line and a cancerous lung cell	2+3	CO3

Course Outcomes

CO1: Students will acquire basic understanding of molecular biology topics: nucleic acid structure and chemistry; organization of genome in chromosomes; regulation of replication, transcription, translation and DNA repair.

CO2: Students will acquire knowledge of recombinant DNA techniques on: nucleic acid amplification and gene cloning; manipulation of DNA sequences; preparation and screening of nucleic acid libraries; gene silencing; analysis of variations in genome sequence.

CO3: Students will be proficient in applying basic understanding of molecular biology topics in analysing and solving problems related to recombinant DNA technology.

CO4: Students will be able to design strategies to solve problems related to recombinant DNA technology.

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
Even Semester End-term Examination, 2025-26

NITDGP/BTECH/Reg/Even/2025-26

Course Code: BTC402

Course Name: Cell Biology and Genetics

Question Paper No.: NITDGP/BTC402/endterm

Full Marks: 60

Time: 90 min

Date of Exam: 22/04/2026

Instructions: Answer all questions

Question No.	Body of the Question	Marks	Mapped CO
1	Answer any 5 questions: a) How is a protein specifically directed to the lysosome? In which organelle is this been determined? b) An individual has a mutation in desmoplakin. What do you think will be the effect on epithelial cells like skin? c) Pinocytosis is a selective intake of primarily fluids and ions, often called cell drinking – Do you agree? Defend. d) Paracrine signalling is..... e) Two membranes A and B are compared. A is more fluid as compared to B. can you suggest two structural differences that contribute to the change in property? f) Which of the following modifications will definitely improve the resolution of a light microscope? - Increasing magnification from 40x to 100x - Using light of shorter wavelength - Decreasing numerical aperture (NA) - Increasing the intensity of illumination g) Antibiotics like chloramphenicol can inhibit protein production in the mitochondria but not in the cytosol- Can you explain why?	2*5	CO2
2	Answer any three questions: a) Name the various stages of mitosis. Cyclin along with CDK..... is required for mitotic entry. What would be the consequence if the cyclin in question do not degrade after mitosis- Would the interacting CDK be still active? What cellular defect would possibly arise as a result? b) Consider the following questions and identify the incorrect one/ ones. Please give your rationale for the incorrect statement/(s) - Cytochrome c release from mitochondria is sufficient to activate executioner caspases in the absence of Apaf-1. - Bax and Bak oligomerization is a critical step in mitochondrial outer membrane permeabilization during intrinsic apoptosis. - Cells in G0 phase lack Cyclin D expression and are therefore incapable of responding to mitogenic signals. - p21-mediated inhibition of CDKs can lead to either reversible cell cycle arrest or entry into senescence depending on cellular context.	3*6	CO3

Course Outcomes

CO1: To understand the basic organization of cells and organisms and the tools needed to study them

CO2: To understand the basic processes of the cell-machinery, cell-cell interaction and the eukaryotic cell cycle.

CO3: To apply the knowledge of cell process regulation and cell cycle in understanding the use of a cell as a biological tool for manufacturing biomolecules.

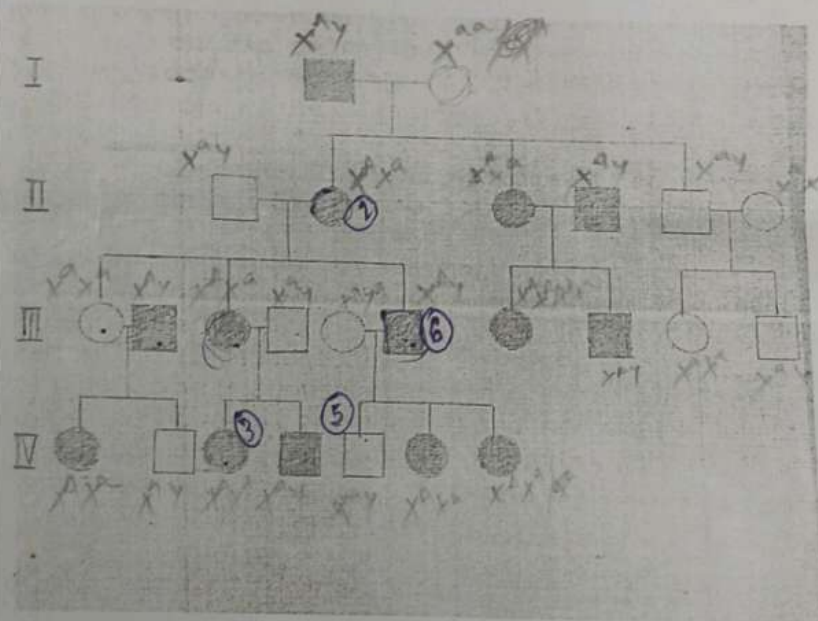
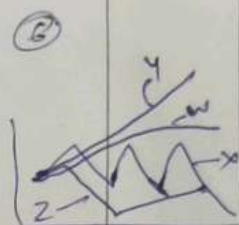
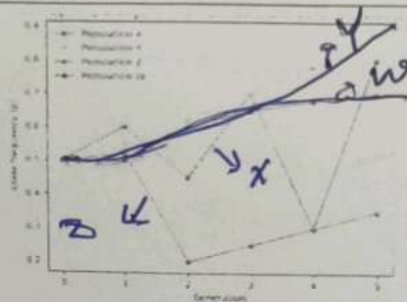
CO4: To learn the fundamentals of Genetics and its applications.

CO5: To solve problems associated with genetic diseases and their transmission from one generation to the next



c) Choose the best option for the Assertion Reason question given below and defend your answer Assertion (A): GPCR signalling usually involves second messengers, whereas RTKs signal only through direct protein-protein interactions. Reason (R): RTKs lack enzymatic activity and depend entirely on adaptor proteins for signalling. Options: A. Both A and R are true, and R is the correct explanation B. Both A and R are true, but R is not the correct explanation C. A is true, R is false D. A is false, R is true What is a second messenger as defined above? Why is it called so? d) Is it true that a proto-oncogene usually adopts a gain of function mutation to become an oncogene? Can you elaborate. A mutation in a signalling protein results in constitutive activation of the MAPK pathway, even in the absence of ligand binding to its receptor. Can you Identify two possible classes of proteins that could be mutated and for each, explain how the mutation may lead to continuous signalling?												
3 Red green colour blindness in humans is considered to be X linked recessive in its inheritance. A woman with phenotypically normal parents and a colour blind brother is married to a colour blind man. What is the probability of the woman to have a colour blind son? Explain	4	CO4										
4. In a plant species, two genes are being studied: Leaf shape: Broad (B) dominant over narrow (b) Flower colour: Red (R) dominant over white (r) A pure-breeding plant with broad red phenotype is crossed with a narrow white plant. The F₁ progeny is test-crossed. a) Make a Punnett Square predicting the phenotypes of the offspring, indicating the ratio. b) However, when the offsprings were analysed (total = 1000); the following result was obtained. <table border="1"><thead><tr><th>Phenotype</th><th>Number</th></tr></thead><tbody><tr><td>Broad Red</td><td>430</td></tr><tr><td>Narrow White</td><td>420</td></tr><tr><td>Broad White</td><td>80</td></tr><tr><td>Narrow Red</td><td>70</td></tr></tbody></table> Can you calculate the recombination frequency between the two genes? Do you think these genes are on the same chromosome? Explain your rationale. c) In a modified analysis using the same plant characteristics; a broad (heterozygous) white plant with is crossed with a double recessive plant. i. Determine the genotypes and phenotypes of the offspring. ii. If there is recombination happening; what would be the outcome in term of phenotype?	Phenotype	Number	Broad Red	430	Narrow White	420	Broad White	80	Narrow Red	70	4+ 4+4	CO4
Phenotype	Number											
Broad Red	430											
Narrow White	420											
Broad White	80											
Narrow Red	70											

5.	The graph alongside shows allele frequency changes across multiple generations for four populations- as a result of genetic drift, bottleneck, gene flow and natural selection. i) Can you define these terms? ii) From the graph, deduce which population corresponds to each of your definitions above. -Explain your rationale	4+4	CO5
6.	A pedigree of inheritable disease is depicted below. a) Assuming the usual depiction of symbols; comment on the genetic inheritance of the disease. Indicate the inheritance pattern of every individual. b) What is the relationship between i) Individual 6, generation III and individual 3, generation IV. ii) Individual 2, generation II and Individual 5, generation IV	6+2	CO5



25
9
38
41

12
25
25
25

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR

Even Semester End-term Examination, 2025-26

Course Code: CHC431

Full Marks: 60

Course Name: Unit Operation of Chemical Engineering (CHC431)

Time: 180 Mins

Question Paper No.: BT/4th Sem/End Sem/CHC431/01

Date of Exam: 18/04/2026

Group-A

Question No.	Body of the Question	Marks	Mapped CO																																													
1	a) What is mesh? Explain the meaning of 200 mesh. Define oversize and undersize in screen operation. b) Particle size distribution data of a screen are presented in the table. Calculate (i) average particle diameter, (ii) cumulative fraction smaller particle (iii) plot the cumulative analysis of smaller particle. <table border="1"> <thead> <tr> <th>Mesh</th><th>Screen Opening, ϕ (mm)</th><th>Mass Retained on Screen, (gm)</th></tr> </thead> <tbody> <tr><td>4</td><td>4.699</td><td>0.000</td></tr> <tr><td>6</td><td>3.327</td><td>0.0251</td></tr> <tr><td>8</td><td>2.362</td><td>0.1250</td></tr> <tr><td>10</td><td>1.651</td><td>0.3207</td></tr> <tr><td>14</td><td>1.168</td><td>0.2570</td></tr> <tr><td>20</td><td>0.833</td><td>0.1590</td></tr> <tr><td>28</td><td>0.589</td><td>0.0538</td></tr> <tr><td>35</td><td>0.417</td><td>0.0210</td></tr> <tr><td>48</td><td>0.295</td><td>0.0102</td></tr> <tr><td>65</td><td>0.208</td><td>0.0077</td></tr> <tr><td>100</td><td>0.147</td><td>0.0058</td></tr> <tr><td>150</td><td>0.104</td><td>0.0041</td></tr> <tr><td>200</td><td>0.074</td><td>0.0031</td></tr> <tr><td>Pan</td><td>-</td><td>0.0075</td></tr> </tbody> </table>	Mesh	Screen Opening, ϕ (mm)	Mass Retained on Screen, (gm)	4	4.699	0.000	6	3.327	0.0251	8	2.362	0.1250	10	1.651	0.3207	14	1.168	0.2570	20	0.833	0.1590	28	0.589	0.0538	35	0.417	0.0210	48	0.295	0.0102	65	0.208	0.0077	100	0.147	0.0058	150	0.104	0.0041	200	0.074	0.0031	Pan	-	0.0075	3+7	CO4, CO5
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2	a) What is the basic difference between crushing and grinding? Explain the role of moisture content in the material for the size reduction. b) A material is crushed in a Blake crusher, and the average particle size is reached from 60 mm to 10 mm with the consumption of energy 13 KW/kg. S. What will be the consumption of energy needed to average particle size from 75 mm to 25 mm? State the reasons.	2+8	CO4, CO5																																													
3	a) What is scaling or fouling during filtration process and how to prevent? Draw a variation of filtration flux with time under washing condition during filtration process b) Data for the laboratory filtration of CaCO_3 slurry in water at 298 K	2+8	CO4, CO5																																													

Course Outcomes

- CO1: To Understand the fundamentals of fluid dynamics and mechanics
 CO2: Understanding the fundamentals of heat transfer operations
 CO3: To learn the design of heat transfer equipment and calculations
 CO4: To develop knowledge of different mechanical operations and their applications
 CO5: To solve related problems of different difficulty levels through tutorials

$$\frac{KW}{kg} = K_R \left(\frac{1}{m} \right)$$

$$K_R = KW / kg \cdot m$$

1

8.6

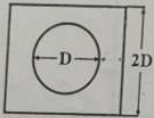
2

7.0

7.7

	are reported as follow at a constant pressure drop of 338 KN/m^2 . The filter area of the plate frame filter press is 0.0439 m^2 & the slurry concentration C is 23.47 kg/m^3 of water. Calculate the constant α & R_m from experimental data:				
	$t \text{ (s)}$	$v \text{ (m}^3\text{)}$	$t/v \text{ (s/m}^3\text{)}$		
	4.4	0.498×10^{-3}	8.84×10^3		
	9.5	1.000×10^{-3}	9.5×10^3		
	16.3	1.501×10^{-3}	10.86×10^3		
	24.6	2.000×10^{-3}	12.3×10^3		
	34.7	2.498×10^{-3}	13.89×10^3		
	46.1	3.002×10^{-3}	15.36×10^3		
	59	3.506×10^{-3}	16.83×10^3		
	73.6	4.004×10^{-3}	18.38×10^3		
	89.4	4.502×10^{-3}	19.86×10^3		
	107.3	5.009×10^{-3}	21.42×10^3		
4	a) What is the terminal velocity for settling of particle? Derive the terminal velocity of spherical particle under laminar zone. What is the purpose of agitation? Define the types of impellers for agitation and mixing process. b) Determine the terminal velocity for a sand particle with an average diameter of 0.5 mm and a density of 2600 kg/m^3 settling in water at 20°C . Density and viscosity values are 998.2 kg/m^3 and $1.002 \times 10^{-3} \text{ N.s/m}^2$.			6+4	CO4, CO5

Group-B

Question No.	Body of the Question	Marks	Mapped CO
1	(a) Define Steady and unsteady state heat transfer. (b) What is the view factor? Calculate the F_{11} , F_{12} , F_{21} , and F_{22} for the following figure.  (c) Why is the steam economy value higher for the multi-effect evaporator? When is the steam economy of the single-effect evaporator equal to 1 or more than 1? (d) An annealing chamber has a composite wall made of a 17 cm thick firebrick layer ($k = 1.1 \text{ W/m}^\circ\text{C}$) and a 13 cm thick ordinary brick layer ($k = 0.70 \text{ W/m}^\circ\text{C}$). The inside and outside wall surface temperatures are 400°C and 45°C , respectively. Calculate the heat loss from 25 m^2 of the furnace wall. Also, determine the temperature between the ordinary brick and the firebrick layers.	1+3+2 +4	CO2
2	(a) Calculate the equivalent diameter of the shell with a square pitch tube layout of 2.54 cm OD of the tube, and the pitch length is 1.25 times the OD. (b) What is a baffle? Why is it used in a shell-and-tube heat exchanger? (c) Draw the schematic diagram of a 1, 4 shell and tube heat exchanger and show the fluid flow path for both tube and shell. (d) Ethanol from the condenser at the top of a distillation column is cooled at a rate of 500 kg/hr from 78°C to 35°C in a counter-current double-pipe	2+2+2 +4	CO3

heat exchanger of hairpin type with an effective length of 15 m. The inner tube is carbon steel, 25 mm ID., and 29 mm OD. The outer pipe ID is 45 mm. Ethanol flows through the annulus. Water, which flows through the inner tube, entering at 20°C and leaving at 30°C, is the coolant.

- Calculate the heat duty of the exchanger.
- Calculate the overall heat transfer coefficient.

- Calculate the heat duty of the exchanger and water flow rate.
 - Calculate the Reynolds and Prandtl numbers for both tube and annulus fluids.
- The following data are given:

The following data are available,
Inner tube ID: 25

Inner tube ID: 25 mm, OD: 29 mm, Wall thickness: 2 mm, thermal conductivity of the tube wall: 74.5 W/m.K.

Outer pipe ID: 45 mm, OD: 498 mm.
Thermo-physical properties:

Thermo-physical properties of ethanol at average temperature (56.5°C)
Specific heat: 2.65 kJ/kg °C, viscosity: 7 cP, density: 789 kg/m³, thermal conductivity: 0.158 W/mK

Specific heat: 2.65 kJ/kg °C, viscosity: 7 cP, density: 760 kg/m³, thermal conductivity: 0.158 W/mK

Specific heat: $4.182 \text{ kJ/kg}^\circ\text{C}$

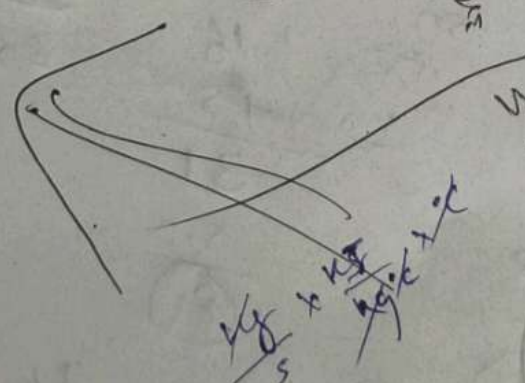
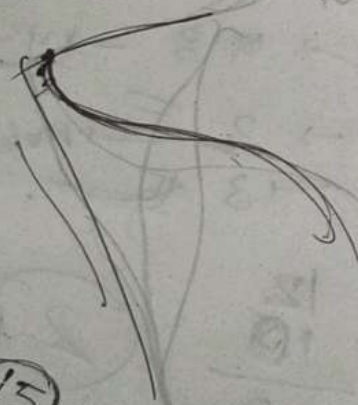
Specific heat: 4.182 kJ/kg °C, viscosity: 0.89 cP, thermal conductivity: 0.607 W/mK, density: 997 kg/m³,
(a) What is the critical Reynolds number?

- What is cavitation?
- Write the effect of temperature on the viscosity of liquid and gas.
- What is NPSH? If the suction pressure is lower than the NPSH, what will happen?
- Write the function and selection criteria of the valve.
- What is the coefficient of discharge? Is the coefficient of discharge of the venturimeter higher than that of the orifice meter? Why?
- Explain the flow measurement mechanism of the Pitot tube with a suitable diagram.

$$1+2+2$$
$$+1+2+\dots+2$$

CO1

2



Raylog
15/4/26

$\frac{K_2}{n_2} \times \frac{K_3}{K_2} \times \frac{K_4}{K_3} \times \frac{K_5}{K_4}$
 $\frac{K_5}{n_5}$

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
Even Semester End-term Examination, 2025-26

NITDGP/BTECH/Reg/Even/2025-26

Course Code: BTC404

Course Name: Immunology

Full Marks: 60

Time: 180 Mins

Instructions: Answer any 6 questions.

Question No.	Body of the Question	Marks	Mapped CO
1	Define Decay Accelerating Factor (DAF). Explain how Factor H and C1-inhibitor regulate the alternative and/or classical pathway. A patient develops recurrent swelling due to deficiency of C1 inhibitor. Explain the immunological basis of this condition.	2+3+5	CO1, CO2
2	What is junctional diversity? What is the function of RAG1 and RAG2? Draw diagram showing RSS and 12/23 rule.	2+3+5	CO1, CO2
3	Explain the T-cell maturation and differentiation. Why are memory cells important in vaccination? Describe opsonization in detail.	4+3+3	CO1, CO2
4	During an immune response, MAC formation increases. What would be the effect on target cells? A patient lacks C3 protein. Explain how this affects complement activation and immunity. Explain the role of invariant chain (Ii) in MHC-II antigen presentation.	3+4+3	CO1, CO2
5	With example explain clinical importance of monoclonal antibodies in therapy. Differentiate between precipitation and agglutination. Blocking is necessary in ELISA. Explain why?	4+3+3	CO3, CO4
6	Describe the role of cytokines in immune regulation and inflammation. Define anaphylaxis. Explain the role of histamine in allergic reactions.	5+2+3	CO2, CO3
7	Identify the sources of proteins for MHC-I presentation. Discuss the role of ADCC (Antibody-Dependent Cellular Cytotoxicity) in Type II reactions. What is a live attenuated vaccine?	4+3+3	CO4, CO5
8	Explain the role of TAP in antigen presentation. Explain the stages of B cell maturation with suitable diagrams.	4+6	CO1, CO2, CO3

BM - TV
DMS
PFS
selection
SPS

Course Outcomes

- CO1: To understand the role of the components of the immune system and its classification
- CO2: To understand the role of the immune cells and their immunological response in the context of human diseases including infectious diseases, autoimmunity, and cancer.
- CO3: To learn the fundamentals and principles of immunological techniques and their application.
- CO4: To understand methods of generations of Polyclonal and Monoclonal Antibody and the use of custom made genetically engineered antibodies.
- CO5: To solve problems associated with drugs and their toxic response based on the knowledge of immunological response.